PROJECT CS214

Pharmacy Inventory Management System

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Section : 6015

INTRODUCTION

This project is a C++ program that manages pharmacy inventory

The system allows users to add, edit, delete, search, and print medicine records

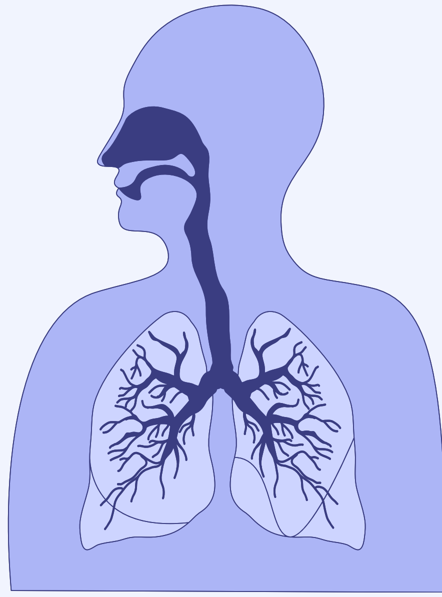
It also stores medicine data in a text file and loads it when the program starts

WHY WE CHOSE THIS PROJECT

We chose the Pharmacy Inventory Management System because it is a simple real-life example that uses data structures clearly

Pharmacies need to manage medicines, search for items quickly, update quantities, and save records

This project helped us apply C++ and data structures in a practical system



PROJECT FUNCTIONS

The system includes the following functions:

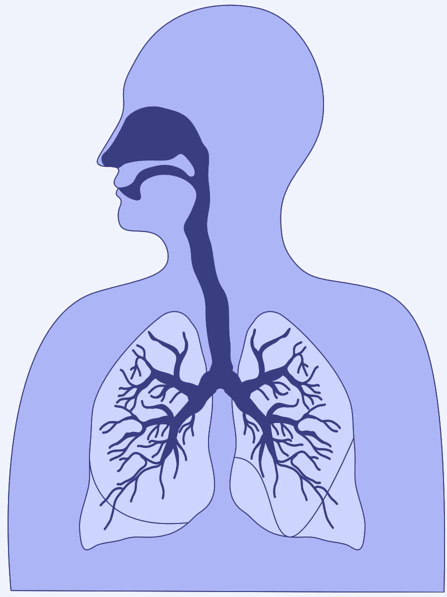
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit

DATA STRUCTURES USED

Array:
Used to store all medicine records with a fixed maximum size

Queue:
Used to display low-stock medicines in first-in-first-out order

Unordered Map:
Used as a hash table to quickly find a medicine by its ID

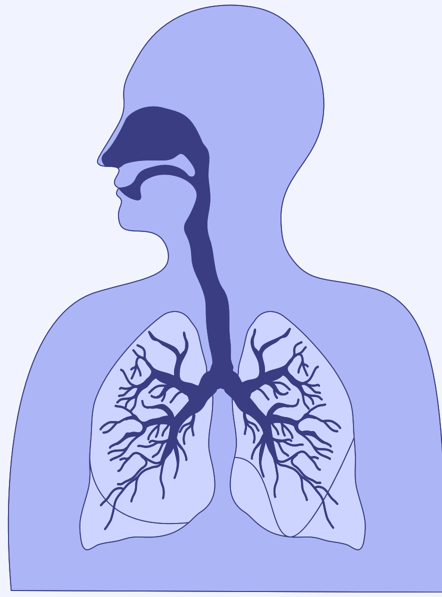


WHY THESE DATA STRUCTURES

Array:
Stores all medicine records in a fixed-size list and allows easy access by index

Queue:
Displays low-stock medicines in first-in-first-out order

Unordered Map:
Improves search by connecting each medicine ID with its index in the array



FLOWCHART

Start
↓
Load data from medicines.txt
↓
Display user menu
↓
User selects an option
↓
If Add / Edit / Delete:
Update data → Save to file → Return to menu
↓
If Search / Print / Low Stock:
Display result → Return to menu
↓
If Exit:
Save data → End

FILE HANDLING

The program uses medicines.txt to store data

When the program starts, it reads medicine records from the file

After adding, editing, or deleting a medicine, the program saves the updated records back to the file

PROJECT LIMITATIONS

- **It stores a limited number of medicines because it uses a fixed-size array**
- **It works only from the console**
- **It searches medicines by ID only**
- **It does not support user login or permissions**
- **It does not check medicine expiry dates**
- **The data is saved in a text file, not a database**

FUTURE IMPROVEMENTS

- **Adding a graphical user interface**
- **Using a database instead of a text file**
- **Adding expiry date tracking**
- **Adding alerts for low-stock medicines**
- **Allowing search by name or category**
- **Adding user login for pharmacists and admins**
- **Generating inventory reports**

RUNNING THE CODE
Added

Microsoft Visual Studio Debu

```
No existing file found A new file will be created later

-- Pharmacy Inventory Management System --
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit
Choose an option: 1
Enter medicine ID: 101
Enter medicine name: Panadol
Enter category: Painkiller
Enter quantity: 20
Enter price: 12.5
Medicine added successfully

-- Pharmacy Inventory Management System --
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit
Choose an option: 1
Enter medicine ID: 102
Enter medicine name: Vitamin C
Enter category: Supplement
Enter quantity: 5
Enter price: 25
Medicine added successfully
```

RUNNING THE CODE
Print All

```
-- Pharmacy Inventory Management System --
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit
Choose an option: 5

All Medicines:
----------------------------------
ID: 101
Name: Panadol
Category: Painkiller
Quantity: 20
Price: 12.5
----------------------------------
ID: 102
Name: Vitamin C
Category: Supplement
Quantity: 5
Price: 25
----------------------------------
```

RUNNING THE CODE
Search

```
-- Pharmacy Inventory Management System --
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit
Choose an option: 4
Enter medicine ID to search: 102

Medicine Found:
ID: 102
Name: Vitamin C
Category: Supplement
Quantity: 5
Price: 25
```

RUNNING THE CODE
Edit

```
-- Pharmacy Inventory Management System --
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit
Choose an option: 2
Enter medicine ID to edit: 101
Enter new medicine name: Panadol Extra
Enter new category: Painkiller
Enter new quantity: 8
Enter new price: 13.5
Medicine updated successfully.
```

RUNNING THE CODE
Print Low stock Medicines

```
-- Pharmacy Inventory Management System --
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit
Choose an option: 6

Low Stock Medicines:
----------------------------------
ID: 101
Name: Panadol Extra
Quantity: 8
----------------------------------
ID: 102
Name: Vitamin C
Quantity: 5
----------------------------------
```

RUNNING THE CODE
Delete + Print All

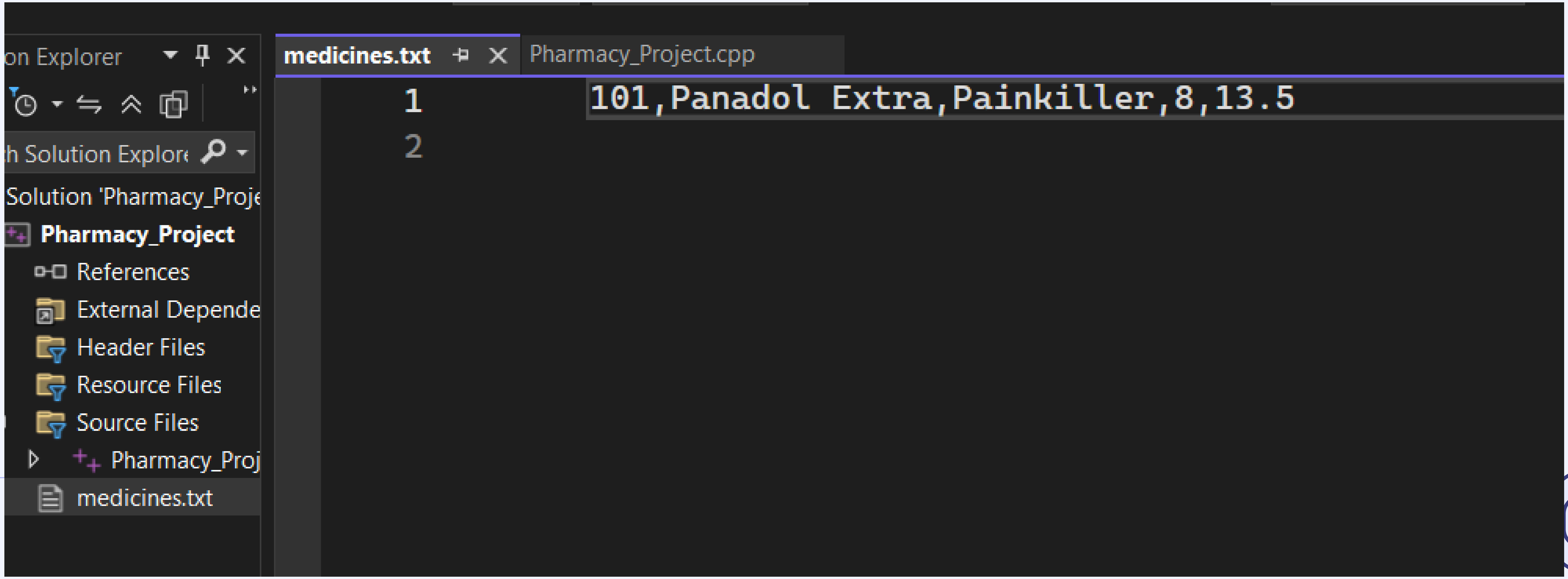
```
-- Pharmacy Inventory Management System --
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit
Choose an option: 3
Enter medicine ID to delete: 102
Medicine deleted successfully

-- Pharmacy Inventory Management System --
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit
Choose an option: 5

All Medicines:
----------------------------------
ID: 101
Name: Panadol Extra
Category: Painkiller
Quantity: 8
Price: 13.5
----------------------------------
```

```
-- Pharmacy Inventory Management System --
1- Add Medicine
2- Edit Medicine
3- Delete Medicine
4- Search Medicine
5- Print All Medicines
6- Print Low Stock Medicines
7- Exit
Choose an option: 7
Data saved Exiting program.
```

medicines.txt : Saved Data File

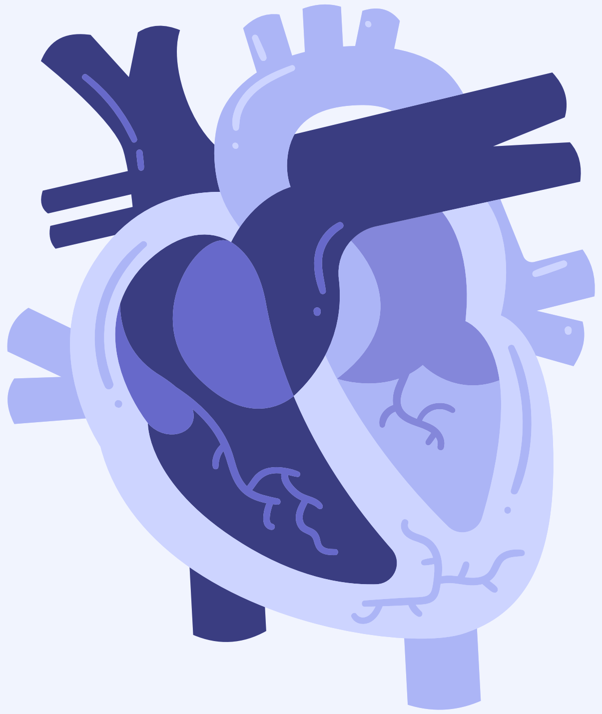


CONCLUSION

This project shows how data structures can be used in a real-life pharmacy inventory system

The array stores medicine records , the queue handles low-stock medicines, and unordered_map improves searching by medicine ID

File handling is used to save the data and load it again after the program exits.



THANK YOU